

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	SWTID-2026-8542
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Story Input	Story prompt entry through web form (index.html) Main character name, setting, tone and art style selection Comic generation through JSON API request (/generate-comic/json)
FR-2	Comic Outline Generation	Structured 5-panel outline using Gemini Flash Panel number, title, scene description and image prompt for each panel
FR-3	Story Narration & Dialogue	Detailed narration for each panel using Gemini Pro Character dialogue generation based on the outline
FR-4	Comic Illustration Generation	Comic-style image for each panel using Stable Diffusion Image saving with sanitized filenames Standalone image test through /test-image
FR-5	Comic Preview & Customization	Panel-by-panel comic preview (comic_preview.html) Regenerate the comic with a different tone or art style
FR-6	Comic Export	Layout building of panel image, title and text PDF generation with timestamped filename Export success confirmation page (/export-success)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple HTML form with clearly labelled fields (prompt, character, setting, tone, art style) and minimal navigation, so non-artists can create a comic in a few steps.
NFR-2	Security	Inputs validated with Pydantic and FastAPI Form parameters; prompts sanitized for safe filenames; Gemini API key kept in environment variables; API calls over HTTPS.
NFR-3	Reliability	AI-call and JSON errors handled with try-except blocks and HTTPException so failures return a clear message instead of crashing the app.
NFR-4	Performance	Gemini Flash used for the fast outline step; Stable Diffusion runs with PyTorch (GPU when available); the model is loaded once and reused for all panels.
NFR-5	Availability	Gemini text generation runs on Google's managed infrastructure; the app runs on Uvicorn and can be run with multiple workers behind a load balancer.
NFR-6	Scalability	Modular layered design (frontend, FastAPI backend, AI modules) with stateless routes, so components can be scaled or split into separate services.